

Teaching Tech Together

Mental Models



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Estimated time: 40 minutes.

All art from openclipart.org (open license)

Learning Objectives

1. Describe the distinctions between novice, competent, and expert learners in the Benner/Dreyfuss model.
2. Explain what a mental model is and why it doesn't need to be either complete or correct in order to be useful.
3. Explain how the lack of a mental model impedes novices, why the goal of lessons for novices should therefore be to help them construct a mental model, and how an instructor can tell if someone has a functional mental model.
4. Explain what a concept map is and what elements a concept map should have.
5. Describe four ways that concept maps can be used in planning, delivering, and assessing lessons.
6. Create concept maps.
7. Define the term "false beginner", explain it in terms of mental models, and explain the special challenges that false beginners present in teaching.

Cognitive Progression



Patricia Benner

- Novice
- Advanced beginner
- Competent
- Proficient
- Expert

Starting in the 1980s, Patricia Benner began studying the way that nurses go from being novices to experts. What she found, and what other researchers have found since looking at other groups, is that people go through a fairly predictable series of *cognitive transitions*. They don't just amass more knowledge: they actually start to think about problems in very different ways.

Cognitive Progression



Patricia Benner

- Novice
- ~~Advanced beginner~~
- Competent
- ~~Proficient~~
- Expert

Benner identified five stages, but for our purposes, we're going to simplify the model to three and say that someone is either a novice, competent, or an expert.

Exercise

What are the symptoms of someone being a novice? How do they act or what do they say that makes you classify them that way?



Take a couple of minutes and think of a few symptoms of someone being a novice. Don't worry yet about the "why" - instead, list the things they do or say that leads you to classify them as a novice cook, a novice driver, a novice speaker of Esperanto, or a novice programmer.

Exercise

- Do things by rote
- Ask nonsensical questions
- Use the wrong terms
- Can't tell what's relevant
- Lack of confidence



Here are some common answers. Before we explore them, we need to take one off the table.

Exercise

- Do things by rote
- Ask nonsensical questions
- Use the wrong terms
- Can't tell what's relevant
- ~~Lack of confidence~~

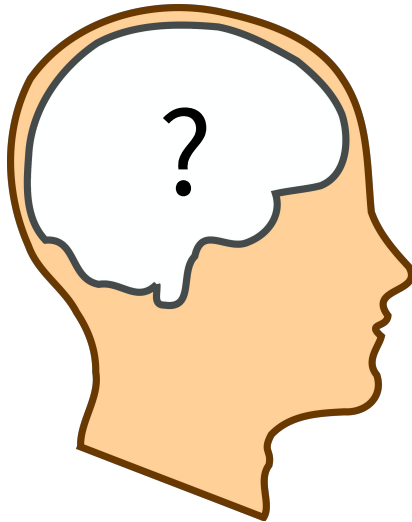


"Let me interrupt your expertise with my confidence."

A lack of confidence doesn't necessarily mean that someone is a novice. People from privileged groups are particularly susceptible to it, which is something we'll return to later.

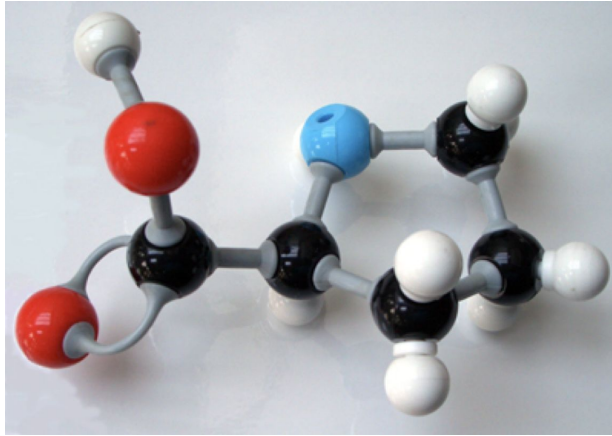
Cartoon from *The New Yorker*, <https://www.newyorker.com/cartoon/a21444>

What Makes a Novice



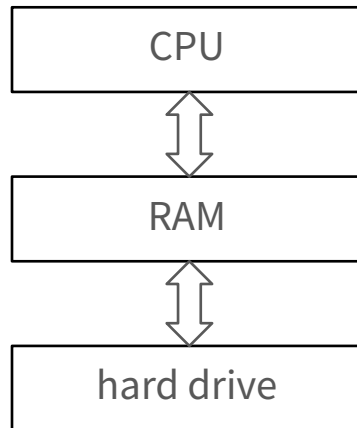
What makes someone a novice is that they don't yet have a *mental model* of the problem: they don't know what the important pieces are or how they relate. They therefore either have to follow steps without understanding why, or borrow concepts from other domains they understand and hope that the analogies hold.

A Mental Model



Here's an example of what we mean by a mental model. In grade school science class, you were probably taught that atoms are brightly colored balls connected by springs. None of those things are actually true, but it's a useful model for understanding the things people want to understand at that stage of their development, and for solving the problems they want to solve.

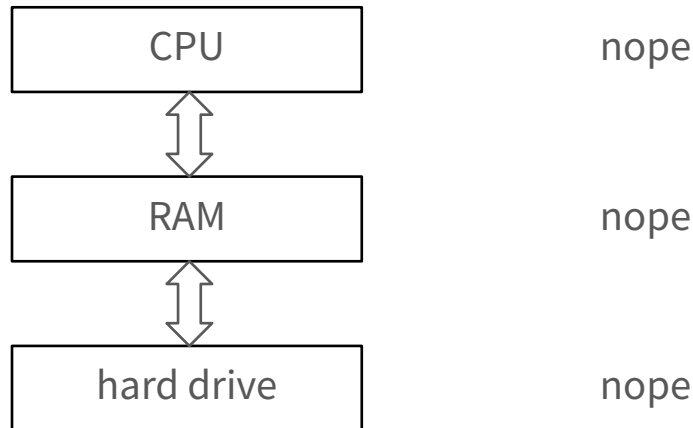
Another Mental Model



Here's another mental model: a computer has a CPU, some RAM, and a hard drive. It's really useful for explaining computer performance to novices...

Descriptive text: the diagram has three boxes. The top box is the CPU, which is connected by a bi-directional arrow to the middle box representing random access memory (labelled RAM), which in turn is connected by a bi-directional arrow to the bottom box labeled "hard drive".

Another Mental Model



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...but again, everything about it is untrue. Your computer probably has several processors, there are at least three levels of memory, and your hard drive probably isn't a drive. None of that matters to most programmers, though: the model allows them to make predictions that they can act on and to fix things that go wrong.

Descriptive text: the diagram has three boxes, each with the word “nope” beside it. The diagram itself is the same as in the previous slide: the top box is the CPU, which is connected by a bi-directional arrow to the middle box representing random access memory (labelled RAM), which in turn is connected by a bi-directional arrow to the bottom box labeled “hard drive”.

How to Help Novices

Help them build a mental model

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Until someone has a mental model, they don't have anywhere to put information, and can't tell which bits are useful when. Your goal when teaching novices is therefore to help them build a mental model. For example, the Software Carpentry introduction to the Unix shell covers 15 commands in 3 hours. That's only one command every 12 minutes, which sounds really slow until you realize the goal is actually to help them understand:

- What's a path?
- What's the difference between an absolute path and a relative path?
- What's a home directory?
- What are command flags?
- What's a pipe?
- What's history and how do I use it?
- What's a wildcard?

Until people know this, nothing else will make sense.

How to Tell When You're Done

*Do they search for the right things?
Can they recognize useful answers?*

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You can tell when people have a usable mental model when they can read the manual—or these days, when they can type the right search into their browser and recognize a useful answer when it pops up.

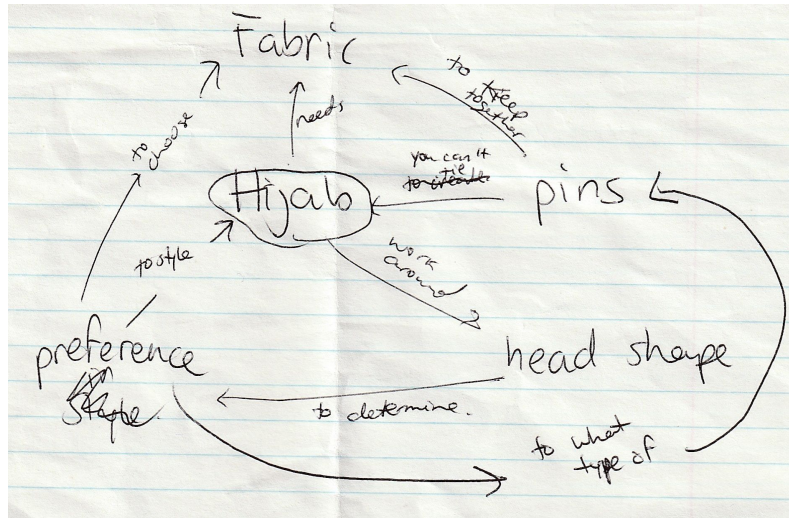
How to Start

Concept maps

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So how do you do this? The first step for any lesson is to figure out what mental model you want them to have, and the best way to do that is to draw a *concept map*.

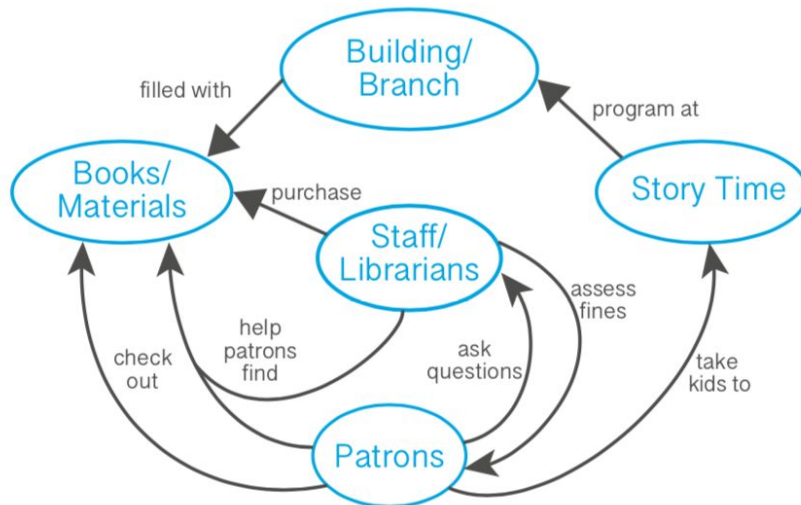
How to Tie a Hijab



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Here's a concept map a learner drew showing how to tie a hijab. Do you have a round face or a long one? Is the fabric stiff or soft? It isn't complete—the class had quite a lively discussion about curly hair versus flat hair—but this is a quick way to understand how someone thinks about something.

Patron's View of a Library

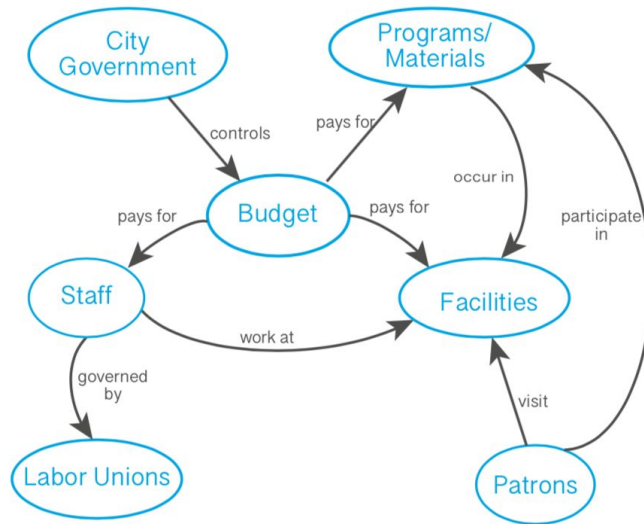


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Here's a patron's view of a library (as drawn by Amy Hodge at Stanford University).

Descriptive text: a graph with five nodes labeled "staff/librarians", "books/materials", "patrons", "story time", and "building/branch". Each node is connected to at least two others, and some pairs of nodes have multiple connections.

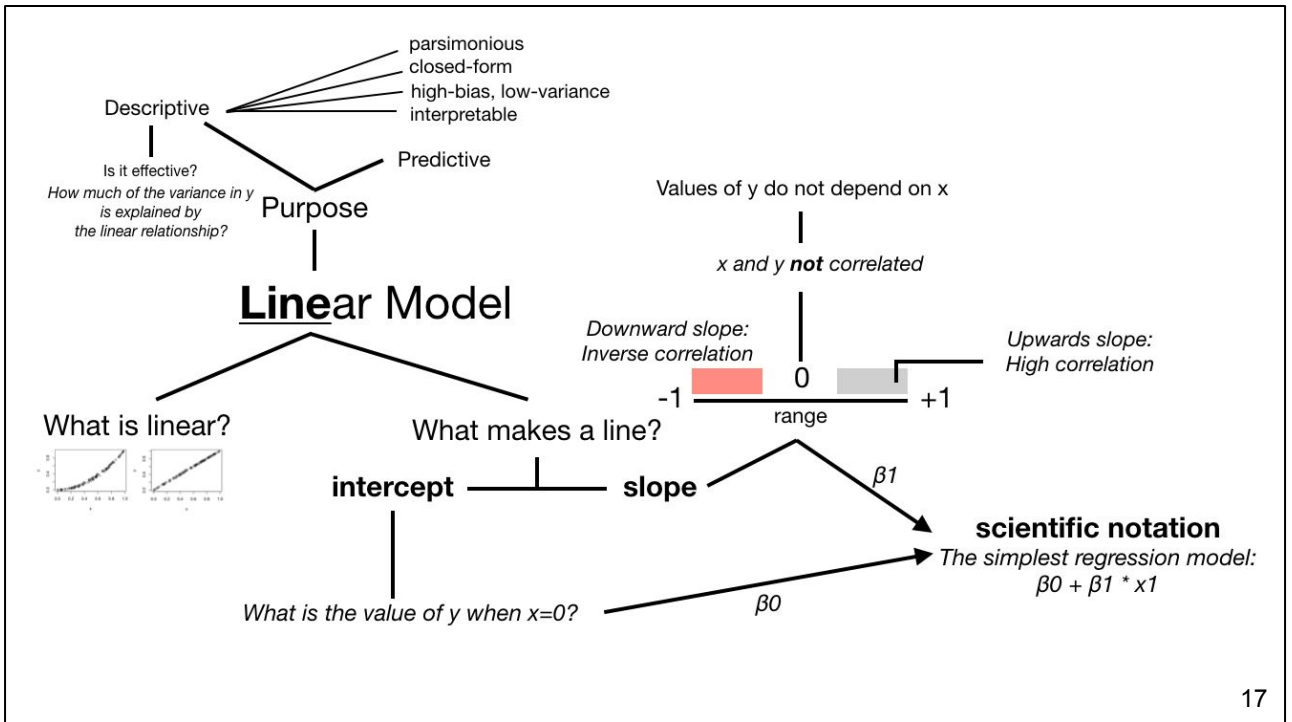
Director's View of a Library



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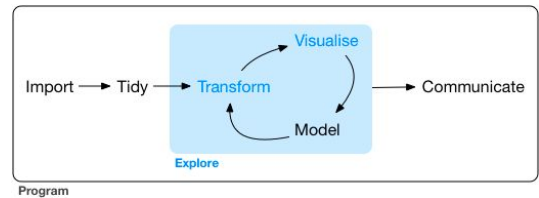
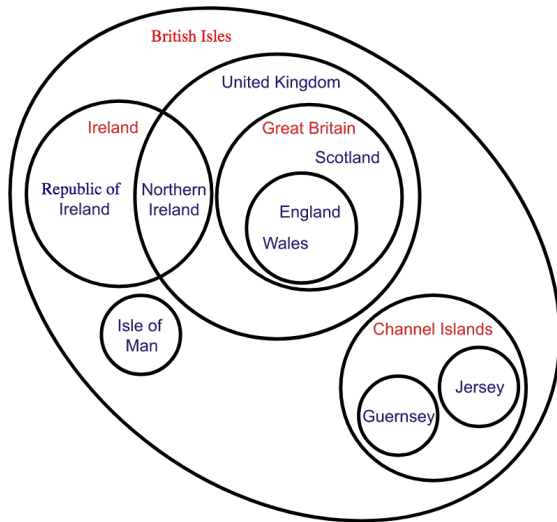
And here's a library director's mental model.

Descriptive text: a graph with seven nodes labeled "budget" (which is in the center), "staff", "facilities", "labor unions", "patrons", "programs/materials", and "city government". Each node is connected to at least two others, and some pairs of nodes have multiple connections.



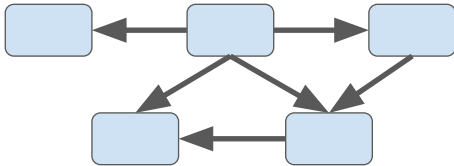
...and sometimes a Venn diagram or flowchart is more accurate. What concept maps do that most other notations don't is *label the relationships*. Those links are how we get from the knowledge we have to the knowledge we need; if you don't tell learners how two facts are connected, they're probably not going to be able to recall things.

Alternatives



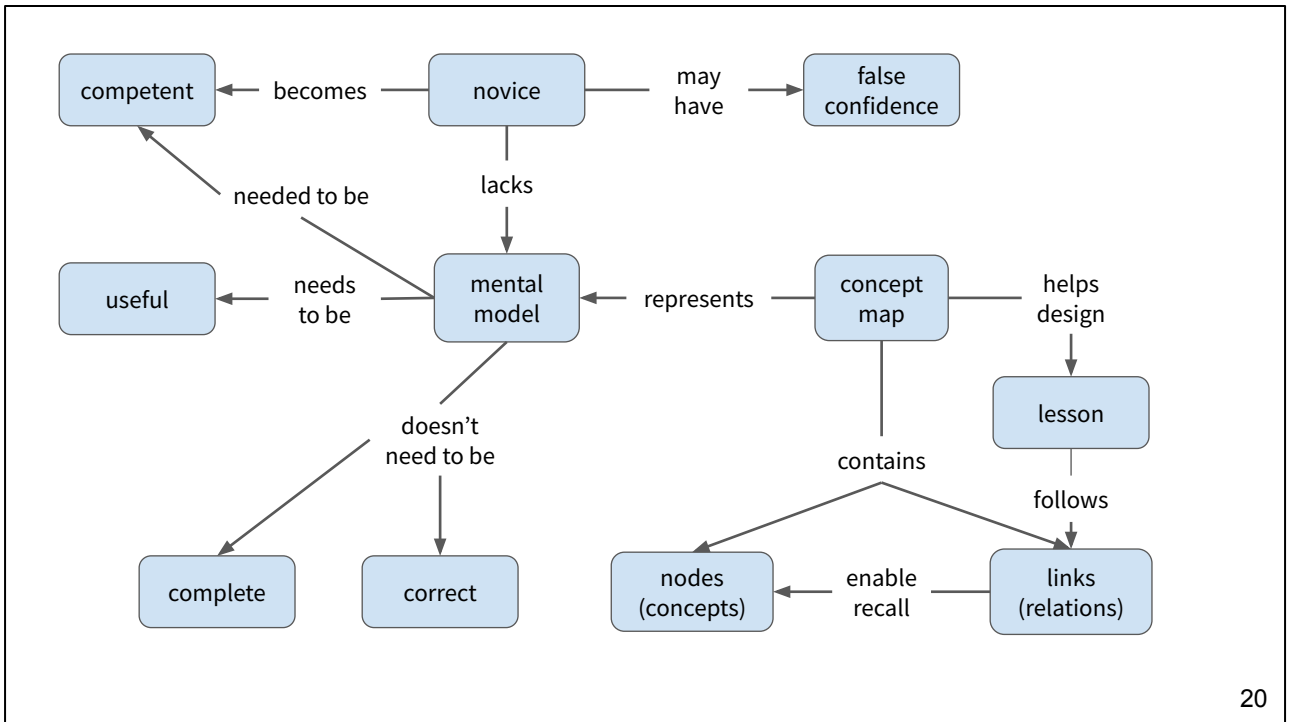
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Use in Lesson Design



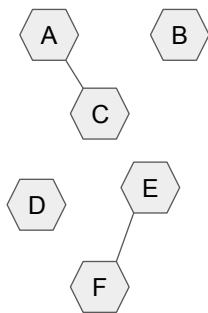
- Define lesson goals
- Separate content and order
- Emphasize relationships
- Communicate with fellow instructors

Drawing a concept map before you start writing a lesson helps you define your lesson's goals. It also helps you focus on content before worrying about order, and reminds you emphasize the relationships between things as well as the things themselves. Finally, it's a good way to communicate with your fellow instructors: the odds are that if you and someone else have very different mental models of a subject, you'll have a hard time teaching it together. We will look later at how to use concept maps when delivering and assessing a lesson.



This is the concept map for this lesson. I drew it *before* creating the lesson to serve as a guide - we'll talk about this later.

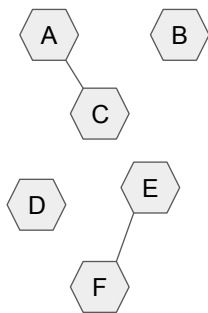
Descriptive text: a concept map summarizing this lesson.



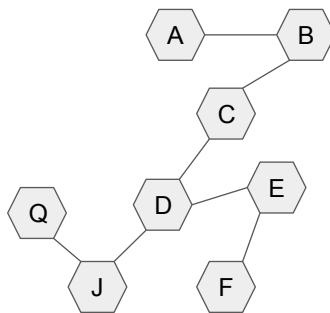
Novice

Concept maps also help explain the differences between novices, competent practitioners, and experts. Novices don't have a lot of connections between concepts.

Descriptive text: on the left of the page is a small graph titled "Novice" with six nodes labeled "A" to "F". "A" and "C" are connected, and "E" and "F" are connected, but those pairs do not connect to each other, while "B" and "D" are not connected at all.



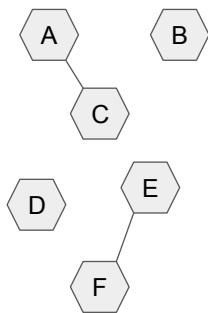
Novice



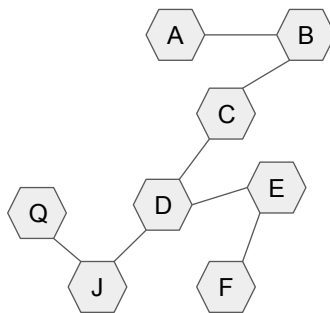
Competent

Competent practitioners have more, but it can take them several steps to get from what they know to what they need.

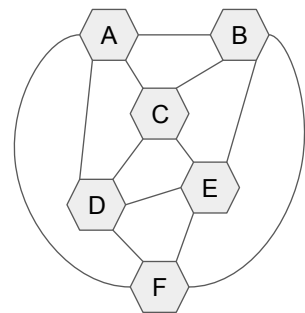
Descriptive text: the six-node “Novice” diagram from the previous slide is repeated on the left. In the center is a slightly larger graph labeled “Competent”. The nodes “A” through “F” are repeated, but now they all join up in order “A” to “B”, “B” to “C”, and so on to “F”. Two nodes “J” and “Q” branch off from node “D”.



Novice



Competent



Expert

Experts have many more connections, so their path is shorter—so short, in fact, that they can often jump from start to finish in a single step. This is also what leads to *expert blind spot*: they can't explain their reasoning because there isn't any.

Descriptive text: the “Novice” and “Competent” diagrams from the previous slide are repeated. On the right, a third graph titled “Expert” has been added. It contains the nodes “A” through “F” with many more connections: “A” is directly connected to “F”, “C” connects to “D” and “E”, and so on.

The Problem of False Beginners



Pre-assessment can give a misleading signal because of *false beginners*, who test at the same level as complete newcomers but will learn more quickly because of prior knowledge.

Exercise

Draw a concept map for something you can explain in 2 minutes. Compare with a neighbor, then share with the class.

- Playing a board game
- How diseases spread
- Tabular data
- Photosynthesis



Take a few minutes to draw a concept map for one of these topics (or something else if you really want to). When you're done, go through your concept map with one of your neighbors and give them feedback on theirs. After that, share your concept maps with the class.